

System Architecture & Tech Stack

1. System Overview

The platform is built on a microservices architecture hosted on AWS. It uses an API Gateway to route traffic to specific domain services.

2. Core Services

User Service: Manages authentication and profiles. Built with Go and backed by PostgreSQL.

Data Processing Service: Handles asynchronous jobs. Built with Python and uses Celery/RabbitMQ.

Reporting Service: Aggregates data for analytics. Built with Node.js and backed by MongoDB.

3. Infrastructure Setup

We use Terraform for Infrastructure as Code (IaC). Services are deployed in Docker containers managed by Kubernetes (EKS).

CI/CD is handled by GitHub Actions, with automated testing required before deployment to staging.